

NEMURO, Japan

WMO: 474200

Lat: 43.3315N

Lon: 145.5860E

Elev: 27

StdP: 101.00

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.4	-9.0	-16.9	0.9	-7.9	-15.5	1.0	-7.4	17.6	-2.4	15.5	-3.8	4.8	240	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	24.7	20.7	22.8	19.9	21.4	19.1	21.7	23.6	20.7	22.2	19.7	21.0	5.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	15.8	22.9	20.1	14.9	21.7	19.1	13.9	20.8	63.7	23.8	60.2	22.3	56.6	21.0	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.4	10.9	-12.6	28.7	2.0	2.3	-14.0	30.3	-15.2	31.6	-16.3	32.9	-17.8	34.6		
(5)			-13.2	23.3	1.9	1.4	-14.6	24.3	-15.8	25.1	-16.9	25.9	-18.3	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-3.8	-4.2	-0.9	3.8	8.0	11.2	15.3	17.8	16.5	11.6	5.4	-0.7	(6)
(7)		DBStd	8.11	2.65	2.88	2.74	2.87	3.24	2.93	3.12	2.94	2.48	2.68	3.43	2.97	(7)
(8)		HDD10.0	1938	428	397	337	189	81	19	1	0	0	14	141	332	(8)
(9)		HDD18.3	4287	686	630	596	437	321	213	106	47	66	208	387	591	(9)
(10)		CDD10.0	744	0	0	0	1	18	56	164	241	196	64	4	0	(10)
(11)		CDD18.3	52	0	0	0	0	0	1	10	29	12	0	0	0	(11)
(12)		CDH23.3	128	0	0	0	0	2	4	34	72	16	0	0	0	(12)
(13)		CDH26.7	17	0	0	0	0	0	1	4	11	1	0	0	0	(13)
(14)	Wind	WSAvg	5.3	6.1	5.7	5.6	5.5	5.3	4.5	4.0	4.3	4.8	5.7	6.1	6.2	(14)
(15)	Precipitation	PrecAvg	1055	41	29	57	76	102	104	112	121	152	121	84	56	(15)
(16)		PrecMax	1622	140	81	126	221	266	263	337	500	427	312	230	173	(16)
(17)		PrecMin	746	6	3	6	9	31	24	13	19	34	32	2	7	(17)
(18)		PrecStd	197	28	19	31	39	52	52	65	80	71	60	46	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.5	8.6	16.5	20.7	22.7	26.7	28.1	25.3	20.0	14.8	9.3	(19)
(20)			MCWB	3.2	2.9	4.6	9.3	13.5	16.9	21.4	22.9	21.1	15.7	12.3	8.0	(20)
(21)		2%	DB	2.6	2.5	5.8	12.5	17.4	19.5	23.7	25.2	23.1	17.8	13.4	7.3	(21)
(22)			MCWB	1.4	1.1	3.5	7.4	12.0	15.4	20.3	21.2	20.2	14.8	11.3	6.0	(22)
(23)		5%	DB	1.3	1.3	4.3	9.9	15.0	17.4	21.6	23.3	21.6	16.6	12.1	5.6	(23)
(24)			MCWB	-0.1	-0.1	2.3	6.3	10.8	14.5	19.2	20.3	19.2	14.2	10.3	4.2	(24)
(25)		10%	DB	0.4	0.2	3.0	7.9	12.9	15.6	19.8	21.7	20.2	15.5	10.8	4.0	(25)
(26)			MCWB	-1.0	-1.0	1.3	5.3	9.8	13.5	18.1	19.8	18.0	13.3	9.0	2.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.3	5.6	10.2	14.7	17.7	22.4	23.4	22.9	17.5	13.4	8.5	(27)
(28)			MCDWB	4.1	4.2	7.4	15.3	18.2	21.0	25.6	27.2	24.2	18.5	14.1	9.1	(28)
(29)		2%	WB	1.7	1.4	3.9	8.1	12.7	16.1	20.9	22.0	21.1	15.8	11.8	6.2	(29)
(30)			MCDWB	2.5	2.2	5.4	11.2	16.2	18.6	23.1	23.9	22.2	16.9	12.8	7.0	(30)
(31)		5%	WB	0.3	0.2	2.7	6.8	11.4	14.8	19.5	21.1	19.8	14.7	10.6	4.3	(31)
(32)			MCDWB	1.1	0.9	4.0	9.0	14.2	16.8	21.1	22.6	20.9	15.9	11.8	5.4	(32)
(33)		10%	WB	-0.8	-0.9	1.5	5.7	10.3	13.6	18.3	20.2	18.5	13.8	9.3	2.6	(33)
(34)			MCDWB	0.2	0.1	2.8	7.6	12.4	15.3	19.7	21.4	19.8	15.1	10.7	3.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	5.2	6.1	6.6	6.1	5.9	5.2	5.0	5.4	5.6	5.1	(35)
(36)			MCDBR	5.9	6.6	7.0	9.8	10.8	9.4	9.0	7.8	6.6	6.3	7.0	7.2	(36)
(37)			MCWBR	5.9	5.8	5.4	6.1	6.4	6.0	5.9	4.6	4.1	4.9	6.5	7.5	(37)
(38)		5% WB	MCDWR	6.1	6.1	6.7	8.9	9.7	8.6	8.6	6.9	5.8	5.4	6.7	7.0	(38)
(39)			MCWBR	6.4	5.8	5.6	6.1	6.3	5.9	5.9	4.6	4.3	5.0	6.8	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.374	0.447	0.508	0.523	0.492	0.475	0.461	0.414	0.400	0.377	0.346	(40)	
(41)		taud	2.220	2.086	1.943	1.860	1.910	2.097	2.216	2.273	2.353	2.244	2.205	2.191	(41)	
(42)		Ebn at Noon	771	795	777	762	765	792	801	800	812	763	709	709	(42)	
(43)		Edh at Noon	92	125	164	193	189	158	138	127	109	106	94	87	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.74	2.62	3.65	4.52	4.69	4.65	4.29	3.90	3.52	2.79	1.84	1.47	(44)	
(45)		RadStd	0.13	0.14	0.27	0.37	0.41	0.43	0.38	0.42	0.30	0.16	0.14	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.32	N/A	N/A	N/A	N/A	N/A	N/A	-109	N/A	N/A
(47) Regional (4 neighbors)		+0.30	N/A	N/A	N/A	N/A	N/A	N/A	-83	N/A	N/A

Nomenclature: See separate page